

08 · Did the loyalty rollout work? — difference-in-differences (CausalPy)

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Runs in the legacy environment (pymc<6): `make env-legacy, kernel cmp-legacy`.

The business decision. We launched the loyalty app in some stores, not others. Revenue in the launch stores went up — but revenue went up *everywhere* (a seasonal tide lifts all boats). Did the app add anything **net of that tide**, and is it worth rolling out to the whole chain?

1.0.1 The idea: two differences

Difference-in-differences (DiD) is the workhorse of quasi-experiments. It takes **two** differences and subtracts them:

$$\text{DiD} = \underbrace{(\bar{Y}_{\text{after}}^{\text{treated}} - \bar{Y}_{\text{before}}^{\text{treated}})}_{\text{treated change}} - \underbrace{(\bar{Y}_{\text{after}}^{\text{control}} - \bar{Y}_{\text{before}}^{\text{control}})}_{\text{control change}}.$$

The first difference (treated after — before) contains the app effect **plus** the seasonal tide; the second (control after — before) is the tide **alone**. Subtracting removes the tide and leaves the app effect. It’s the same “compare like with like” logic as everywhere else, applied *across time*.

1.0.2 The one assumption everything rests on: parallel trends

DiD is valid only if, **absent the app**, treated and control stores would have moved *in parallel* — the control’s change is a fair stand-in for what the treated stores’ change would have been. This is untestable after launch (we never see the treated stores’ no-app future), but the **pre-launch trends are the evidence**: if the two groups moved together *before* the app, parallel trends is credible. We check this formally with an **event study** (the effect period-by-period) and falsify with a **placebo** (a fake launch date in the pre-period, which should show ~ 0 effect).

On real data. Swap in your **own store/region panel** — one row per store per period with revenue, a treated/control flag, and a period index. The textbook public example is **Card & Krueger (1994)** on minimum wage and employment (fast-food restaurants in New Jersey vs Pennsylvania). One warning we *demonstrate live* in Step 7: if stores adopt at **different times** (“staggered rollout”), naive DiD can be badly biased and you need modern estimators (Callaway–Sant’Anna, Sun–Abraham).

7-step contract.

1.1 2 · Simulate a ground truth

40 stores, half get the loyalty app at week 12. Every store shares a seasonal pattern and has its own baseline; the app adds a **true €400/store/week**. Treated and control move in **parallel** before launch by construction — so DiD should recover €400, and the event study should show a flat pre-trend.

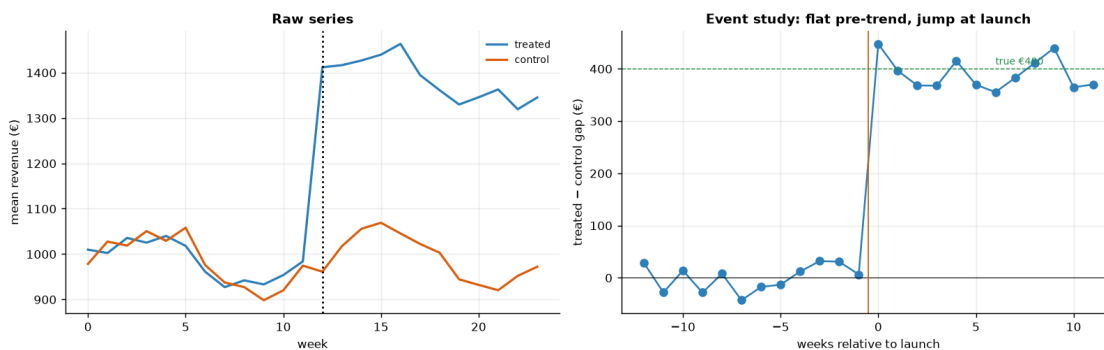
TRUE effect = €400/store/week · 40 stores, 24 weeks, launch week 12

	store	unit	t	week	group	post_treatment	post	revenue
0	store_00	0	0	0	1	False	0	1071.613014
1	store_00	0	1	1	1	False	0	1066.597066
2	store_00	0	2	2	1	False	0	1019.187212
3	store_00	0	3	3	1	False	0	834.246887
4	store_00	0	4	4	1	False	0	1030.278745

1.2 3 · Identify — the 2x2 estimand, parallel trends, and the event study

DiD estimates β_3 in $Y = \beta_0 + \beta_1 \text{group} + \beta_2 \text{post} + \beta_3 (\text{group} \times \text{post}) + \varepsilon$, which equals the difference of the two before/after differences. **Key assumption — parallel trends:** absent the app, treated and control revenue would have moved together. It's untestable post-launch, but the **event study** — the treated-minus-control gap in *each* period relative to launch — is the evidence: flat and ~0 before launch supports it; a jump at launch is the effect.

Pre-launch gaps average €22 (~0 supports parallel trends); post-launch gaps jump toward €400.



1.3 4 · Estimate — Bayesian difference-in-differences

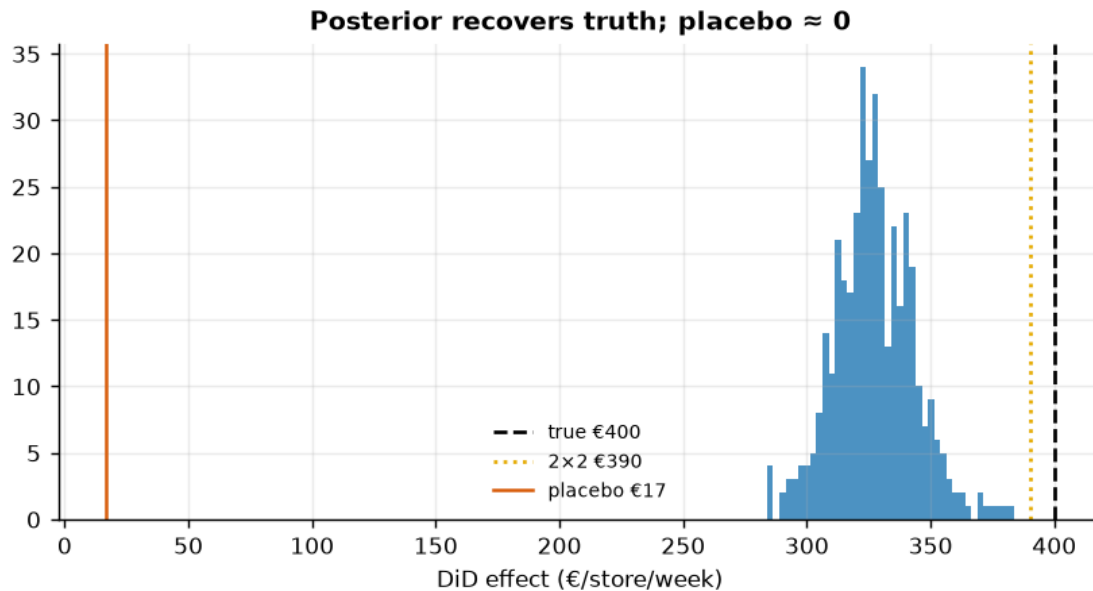
CausalPy's `DifferenceInDifferences` expects the classic **2x2 form** — one pre and one post observation per unit — so we collapse each store's weekly revenue to a pre- and post-launch mean.

DiD effect €327/store/week (true €400) · 90% CI [€302, €353]

1.4 5 · Validate — 2x2 cross-check and a placebo test

Cross-check the Bayesian estimate against the hand-computed 2x2 (the priors mildly shrink it), then **falsify**: run DiD on the *pre-period only*, splitting it into a fake “before/after” at week 6. There was no treatment then, so the placebo effect should be ~ 0 — a large one would mean our design manufactures effects.

Bayesian €327 · 2x2 €390 · true €400 · placebo (no real effect) €17



1.5 6 · Decide, in euros — the rollout case

Net €207/store/week · projected annual value €5,389,807 [90% €4,728,804, €6,069,367]

P(app pays) 1.00 → ROLL OUT

Break-even: the app stays profitable up to a running cost of $\sim$ €302/store/week (5th pct of the effect).

1.6 7 · Caveats — and the staggered-adoption trap, demonstrated

Parallel trends is the whole ballgame and untestable post-launch; a diverging pre-trend kills the design (use synthetic control, nb 07). **No spillover** between nearby stores. But the subtlest trap is **staggered adoption**: when units adopt at *different* times, naive two-way fixed-effects DiD uses already-treated units as controls (“forbidden comparisons”) and can put **negative weights** on some effects (Goodman-Bacon), biasing the estimate even when every true effect is positive. We show it live:

Naive TWFE €177 vs true average post-treatment effect €400 - bias -223. Use Callaway-Sant'Anna / Sun-Abraham for staggered rollouts.

